

The University of Lahore
Department of Computer Science& IT
Final-term Examination



Spring-2023

Course Title:	Discrete Structures	Course Code:	CS-09202	Credit Hours:	3
Course Instructors:	Mr. Tayyab, Miss Azka, Miss Samra, Miss Tehseen	Program Name:	BSCS		
Time Allowed:	120 Minutes	Maximum Marks:	45		
Date:	08-06-2023	Course Mentor: Dr. Mudasser Naseer Signature:			
Student’s Name:					
Reg. No:		Section:			

Important Instructions / Guidelines:

- Understanding of the questions is the part of the question paper so does not disturb the invigilators.
- Kindly leave your mobile phones outside exam hall.
- Solve all questions on the answer sheet provided separately.
- Your paper will be marked as ZERO, if involved in any kind of cheating.

Question # 1 Let $A = \{1,2,3,4\}$ and f and g are functions from A to A such that

$$f(1) = 1, f(2) = 4, f(3) = 1, f(4) = 4 \text{ and } g(1) = 2, g(2) = 3, g(3) = 4, g(4) = 1$$

- a) For each of f and g , determine if it is one-to-one, onto and bijective. [5]

Answer: $f(1)=1$ & $f(3)=1$ hence f is not one-to-one. 2 is not image of any element hence f is not onto and therefore not bijective. g is one-to-one, onto and therefore bijective.

- b) Check if f and g are invertible functions if any of them is invertible find its inverse function.

Answer: f is not invertible as it is not bijective. $g^{-1}(1)=4, g^{-1}(2)=1, g^{-1}(3)=2, g^{-1}(4)=3$. [5]

- c) Give the composite functions $f \circ g$ and $g \circ f$.

Answer: $f \circ g(1) = f(g(1)) = f(2) = 4$	$g \circ f(1) = g(f(1)) = g(1) = 2$
$f \circ g(2) = f(g(2)) = f(3) = 1$	$g \circ f(2) = g(f(2)) = g(4) = 1$
$f \circ g(3) = f(g(3)) = f(4) = 4$	$g \circ f(3) = g(f(3)) = g(1) = 2$
$f \circ g(4) = f(g(4)) = f(1) = 1$	$g \circ f(4) = g(f(4)) = g(4) = 1$

Question # 2 Prove each of the following by applying the asked method:

- a) Prove by contraposition that for an integer n , if n^2 is odd then n is odd. [5]

Answer: Let n is even. Then $n=2k$ (for some int k). Then $n^2=2.2k^2$. Hence, n^2 is even.

- b) Apply mathematical induction to show: $2+4+6+\dots+2n = n(n+1)$ for $n \geq 1$ [5]

Answer: Basic Step: $P(1)$. LHS=1. RHS=1(1+1)=2. So LHS=RHS

Inductive Step: Let $P(k)$ for some k , $2+4+6+\dots+2k = k(k+1)$

$$2+4+6+\dots+2k+2(k+1) = k(k+1) + 2(k+1) \text{ adding } 2(k+1) \text{ on both sides} \\ = (k+1)(k+2) \text{ Hence } P(k+1).$$

Question # 3 Use rules of inference to show:

[8]

If $\forall x (P(x) \vee Q(x))$, $\forall x (\neg Q(x) \vee S(x))$, $\forall x (R(x) \rightarrow \neg S(x))$ and $\exists x \neg P(x)$ are true then $\exists x \neg R(x)$ is true.

Answer:

- | | |
|--|----------------------------|
| 01. $\exists x \neg P(x)$ | Premise |
| 02. $\neg P(c)$ | Existential Instantiation |
| 03. $\forall x (P(x) \vee Q(x))$ | Premise |
| 04. $P(c) \vee Q(c)$ | Universal Instantiation |
| 05. $Q(c)$ | Disjunctive syllogism |
| 06. $\forall x (\neg Q(x) \vee S(x))$ | Premise |
| 07. $\neg Q(c) \vee S(c)$ | Universal Instantiation |
| 08. $S(c)$ | Disjunctive syllogism |
| 09. $\forall x (R(x) \rightarrow \neg S(x))$ | Premise |
| 10. $R(c) \rightarrow \neg S(c)$ | Universal Instantiation |
| 11. $\neg R(c)$ | Modus Tollens |
| 12. $\exists x \neg R(x)$ | Existential Generalization |

Question # 4 Let $F(x)$: x is a female. $P(x)$: x is a parent. $Mother(x,y)$: x is mother of y .

a) Translate from English into logical expression using above predicates and quantifiers. [6]

I. There is a parent who is someone's mother.

Answer: $\exists x, y P(x) \wedge Mother(x, y)$

II. Every female parent is someone's mother.

Answer: $\forall y \exists x F(y) \wedge P(y) \rightarrow Mother(y, x)$

III. There is a female who is no one's mother.

Answer: $\exists x \forall y F(x) \wedge \neg Mother(x, y)$

b) Translate the following logical expression into English. [6]

I. $\forall x \forall y (P(x) \rightarrow (F(x) \vee Mother(x, y)))$

Answer: Every parent is either a female or everyone's mother.

II. $\forall x (\neg P(x) \vee F(x))$

Answer: Every person is either a female or not a parent.

III. $\forall x (F(x) \rightarrow \neg P(x))$

Answer: No female is a parent.